



PES University, Bangalore

(Established Under Karnataka Act 16 of 2013)

Department of Science and Humanities

Engineering Mathematics - I (UE25MA141A)

Classwork problems

Unit - 3: Partial Differential Equations

Form the PDE by eliminating arbitrary functions from the following:

1. $z = f(x + it) + g(x - it)$, where $i = \sqrt{-1}$.

Answer: $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$.

2. $z = x f_1(x + t) + f_2(x + t)$.

Answer: $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial t} + \frac{\partial^2 z}{\partial t^2} = 0$.

3. $f(x + y + z, x^2 + y^2 + z^2) = 0$.

Answer: $(y - z)p + (z - x)q = x - y$.

Solve the following Lagrange's linear PDE of first order: $Pp + Qq = R$, where P, Q, R are the functions of x, y, z .

4. $\frac{y^2 z}{x} p + xzq = y^2$.

Answer: $\phi(x^3 - y^3, x^2 - z^2) = 0$.

5. $pz - qz = z^2 + (x + y)^2$.

Answer: $\phi[x + y, \log(x^2 + y^2 + z^2 + 2xy) - 2x] = 0$.

6. $(mz - ny)p + (nx - lz)q = ly - mx$.

Answer: $\phi(x^2 + y^2 + z^2, lx + my + nz) = 0$.

7. $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$.

Answer: $\phi\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0$.

8. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$.

Answer: $\phi\left(\frac{y}{z}, \frac{x^2 + y^2 + z^2}{z}\right) = 0$.

9. $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$.

Answer: $\phi\left(\frac{x-y}{y-z}, xy + yz + zx\right) = 0$.

10. Solve $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ by the method of separation of variables.

Answer: $u(x, y) = \left[c_1 e^{(1+\sqrt{1-p^2})x} + c_2 e^{(1-\sqrt{1-p^2})x} \right] c_3 e^{p^2 y}$.

11. Obtain the solution of one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, where $c^2 = \frac{k}{sp}$.

Answer: $u = (C_1 \cos px + C_2 \sin px) e^{-c^2 p^2 t}$.

Solve the following higher-order linear equations with constant coefficients.

Strictly follow the “Jain and Iyengar” approach:

The following notations are used.

$$p = \frac{\partial z}{\partial x} = Dz; q = \frac{\partial z}{\partial y} = D'z; D^2z = \frac{\partial^2 z}{\partial x^2}; D'^2z = \frac{\partial^2 z}{\partial y^2}; DD'z = \frac{\partial^2 z}{\partial x \partial y}.$$

12. (i) $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = 0$; (ii) $(D + 2D')(D - 3D')^2z = 0$; (iii) $4r - 12s + 9t = 0$.

Answer: (i) $z = f_1(y + 3x) + f_2(y - 2x)$.

(ii) $z = f_1(y - 2x) + f_2(y + 3x) + xf_3(y + 3x)$.

(iii) $z = \phi_1\left(y + \frac{3}{2}x\right) + x\phi_2\left(y + \frac{3}{2}x\right)$.

13. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$.

Answer: $z = f_1(y - x) + f_2(y + 2x) + xf_3(y + 2x) + \frac{1}{27}e^{x+2y}$.

14. $r - 4s + 4t = e^{2x+y}$.

Answer: $z = f_1(y + 2x) + xf_2(y + 2x) + \frac{x^2}{2}e^{2x+y}$.

15. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} = \sin x \cos 2y$.

Answer: $z = f_1(y) + f_2(y + x) + \frac{1}{2}\sin(x + 2y) - \frac{1}{6}\sin(x - 2y)$.

16. $(D^2 + DD' - 6D'^2)z = \cos(2x + y)$.

Answer: $z = f_1(y + 2x) + f_2(y - 3x) + \frac{x}{5}\sin(2x + y)$.

17. $\frac{\partial^2 z}{\partial x^2} + 3\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 12xy$.

Answer: $z = f_1(y - x) + f_2(y - 2x) + 2x^3y - \frac{3}{2}x^4$.

18. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = y \cos x$.

Answer: $z = f_1(y - 3x) + f_2(y + 2x) - y \cos x + \sin x$.

19. $(2D^2 + 5DD' + 3(D')^2)z = ye^x$.

Answer: $z = f_1(3x - 2y) + f_2(x - y) + \frac{1}{4}(2y - 5)e^x$.

20. $(D^2 + DD' - 2(D')^2)z = 8 \cdot \ln(x + 5y)$.

Answer: $z = f_1(2x - y) + f_2(x + y) - \left[4x^2 + \frac{1}{22}(x + 5y)^2 [2 \cdot \ln(x + 5y) - 1]\right]$.

Unit - 4: Special Functions

Beta and Gamma functions:

Prove the following standard results.

1. $\beta(m, n) = \beta(n, m)$.
2. $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta d\theta = 2 \int_0^{\frac{\pi}{2}} \sin^{2n-1}\theta \cdot \cos^{2m-1}\theta d\theta$.
3. $\int_0^{\frac{\pi}{2}} \sin^p\theta \cdot \cos^q\theta d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right), p > -1, q > -1$.
4. $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$.
5. Relationship between Beta and Gamma functions: Prove that

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}, m > 0; n > 0.$$

6. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.
7. Given that $\int_0^\infty \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin p\pi}$, show that $\Gamma(p) \cdot \Gamma(1-p) = \frac{\pi}{\sin p\pi}, 0 < p < 1$.

Problems on Beta and Gamma functions:

1. Evaluate $\int_0^\infty \sqrt{x} \cdot e^{-x^2} dx$. **Answer:** $\frac{1}{2}\Gamma\left(\frac{3}{4}\right)$
2. Evaluate $\int_0^\infty e^{-x^3} dx$. **Answer:** $\frac{1}{3}\Gamma\left(\frac{1}{3}\right)$
3. Using Beta and Gamma functions, evaluate the integral $\int_{-1}^1 (1-x^2)^n dx$, where n is a positive integer. **Answer:** $\frac{2^{2n+1}(n!)^2}{(2n+1)!}$
4. Evaluate $\int_0^\infty (x^2 + 4)e^{-2x^2} dx$. **Answer:** $\frac{17\sqrt{\pi}}{8\sqrt{2}}$
5. Evaluate $\int_0^\infty 2^{-9x^2} dx$ using the Gamma function. **Answer:** $\frac{1}{6}\sqrt{\frac{\pi}{\ln 2}}$
6. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\sin^8 x}}{\sqrt{\cos x}} dx$. **Answer:** $\frac{60}{13} \frac{\Gamma(\frac{5}{6}) \cdot \Gamma(\frac{1}{4})}{\Gamma(\frac{1}{12})}$

7. $\int_0^2 (8 - x^3)^{-\frac{1}{3}} dx$. **Answer:** $\frac{2\pi}{3\sqrt{3}}$
8. $\int_0^a x^4 \sqrt{a^2 - x^2} dx$. **Answer:** $\frac{\pi a^6}{32}$
9. Evaluate $\int_0^3 \frac{x^{\frac{3}{2}}}{\sqrt{3-x}} \times \int_0^1 \frac{dx}{\sqrt{1-x^{\frac{1}{4}}}}$. **Answer:** $\frac{432\pi}{35}$
10. Evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\cos\theta + \sin\theta)^{\frac{1}{3}} d\theta$. **Answer:** $\frac{6\sqrt{\pi}}{2^{\frac{5}{6}}} \frac{\Gamma(\frac{2}{3})}{\Gamma(\frac{1}{6})}$

Special Functions:

1. Bessel's Differential Equation and Bessel Functions: Obtain the series solution of Bessel's differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - v^2)y = 0$$

of order v , where v is a non-negative real number.

2. Prove that $J_{-n}(x) = (-1)^n J_n(x)$, where n is a positive integer.

Recurrence relations (No proof)

3. $[x^v J_v(x)]' = x^v J_{v-1}(x)$.
4. $[x^{-v} J_v(x)]' = -x^{-v} J_{v+1}(x)$.
5. $x J_v'(x) = x J_{v-1}(x) - v J_v(x)$.
6. $x J_v'(x) = v J_v(x) - x J_{v+1}(x)$.
7. $2 J_v'(x) = J_{v-1}(x) - J_{v+1}(x)$.
8. $2v J_v(x) = x [J_{v-1}(x) + J_{v+1}(x)]$.

Problems:

9. Express $J_5(x)$ in terms of J_0 and J_1
10. Express $J_{\frac{5}{2}}(x)$ and $J_{-\frac{5}{2}}(x)$ in terms sine and cosine functions.
11. Evaluate (i) $J_3(x)dx$ (ii) $x^4 J_1(x)dx$

12. **Generating function for Bessel function $J_n(x)$:** Obtain the generating function for Bessel's function as $e^{\frac{1}{2}x(t-\frac{1}{t})}$.
13. Establish the **Jacobi series**:
- i) $\cos(x\cos\theta) = J_0 - 2J_1\cos 2\theta + 2J_4\cos 4\theta - \dots$
 - ii) $\sin(x\cos\theta) = 2[J_1\cos\theta - J_3\cos 3\theta + J_5\cos 5\theta - \dots]$.
14. Prove that $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x\sin\theta)d\theta$, where n is a positive integer.
This is called **Bessel's integral formula**.
15. Prove that $J_0^2 + 2J_1^2 + 2J_2^2 + 2J_3^2 + \dots = 1$.